STA 431 Midterm II

FRONT: FORMULAS + SETUP

Handwritten-copy master • US Letter landscape • Back: calculator workflows with no downloaded program

1 Joint laws, support, probabilities

Normalize. PMF $p \geq 0$, $\sum\sum p = 1$; PDF $f \geq 0$, $\iint_S f = 1$. If $f = cq$, then $c = 1/\iint_S q$ (sum for PMF). State S; zero elsewhere.

$$p_X(x) = \sum_y p(x,y), \quad f_X(x) = \int_{y:(x,y)\in S} f(x,y)\,dy.$$

Eliminate the other variable; the marginal support is the projection of S. Swap x, y for Y.

$$p_{X|Y}(x \mid y) = \frac{p(x,y)}{p_Y(y)}, \quad f_{X|Y}(x \mid y) = \frac{f(x,y)}{f_Y(y)}.$$

Denominator positive; keep the slice of S at the given y; normalize over x. $f(x,y) = f_{X|Y}(x \mid y) f_Y(y)$. Two marginals alone, or one conditional alone, generally do not determine the joint.

CDF. $F(x,y) = P(X \leq x, Y \leq y)$.

$$F(x,y) = \sum_{u\leq x}\sum_{v\leq y} p(u,v) \quad \text{or} \quad \int_{-\infty}^{x}\int_{-\infty}^{y} f(u,v)\,dv\,du$$

(use the applicable sum or integral). Intersect with S.

$$F_X(x) = \lim_{y\to\infty} F(x,y), \quad f(x,y) = \frac{\partial^2 F}{\partial x \partial y} \text{ a.e.}$$

The derivative formula needs an absolutely continuous law.

$$P(a < X \leq b, c < Y \leq d) = F(b,d) - F(a,d) - F(b,c) + F(a,c).$$

For integer inclusive lower endpoints use $a-1, c-1$; otherwise left limits. Integer PMF $= F(x,y) - F(x-1,y) - F(x,y-1) + F(x-1,y-1)$.

Event bounds. $P((X,Y) \in A) = \iint_{S\cap A} f$ or sum allowed pairs. Draw the intersection and split where a boundary changes. Inner limits may depend on the outer variable. On $0 < x < 1, 0 < y < 2$,

$$P(X < Y) = \int_0^1 \int_x^2 f\,dy\,dx.$$

On $x, y > 0, x + y < 1$, use $0 < x < 1$, $0 < y < 1 - x$. For $X + Y \leq a$, intersect $y \leq a - x$ with S.

Words. Both survive t: $X > t, Y > t$. Series fails if either fails: $T = \min(X,Y)$, $P(T \leq t) = 1 - P(X > t, Y > t)$. Parallel fails if both fail: $T = \max(X,Y)$, $P(T \leq t) = F(t,t)$.

Independence. $F(x,y) = F_X(x)F_Y(y)$ for all x, y; equivalent to $p = p_X p_Y$ at every pair, or $f = f_X f_Y$ almost everywhere, including zero regions. A point failure disproves discrete independence; use positive-area failure for PDFs. Product formula on triangular support is insufficient.

2 Expectation, moments, MGFs

LOTUS.

$$\mathrm{E}[g(X)] = \sum_x g(x)p_X(x) \quad \text{or} \quad \int g(x)f_X(x)\,dx.$$

For $g(X,Y)$ use the joint sum/integral. A finite expectation needs $\mathrm{E}\,|g| < \infty$.

$$\mathrm{E}(aX + bY + c) = a\,\mathrm{E}\,X + b\,\mathrm{E}\,Y + c.$$

Linearity needs no independence. Generally $\mathrm{E}\,g(X) \neq g(\mathrm{E}\,X)$; $\mathrm{E}\,1_A = P(A)$.

Raw and central moments.

$$\mu'_r = \mathrm{E}\,X^r, \quad \mu = \mu'_1, \quad \mu_r = \mathrm{E}[(X-\mu)^r].$$

$$\mu'_0 = 1, \quad \mu_1 = 0, \quad \mathrm{Var}\,X = \mu_2 = \mu'_2 - \mu^2.$$

$$\mu_3 = \mu'_3 - 3\mu\mu'_2 + 2\mu^3,$$

$$\mu_4 = \mu'_4 - 4\mu\mu'_3 + 6\mu^2\mu'_2 - 3\mu^4.$$

$$\mathrm{Var}(aX + b) = a^2\,\mathrm{Var}\,X, \quad \sigma_{aX+b} = |a|\sigma_X.$$

$Z = (X - \mu)/\sigma$ has mean 0, variance 1; not necessarily normal. Symmetry about zero makes existing odd moments vanish. For $Y = g(X)$, $\mathrm{Var}\,Y = \mathrm{E}[g(X)^2] - (\mathrm{E}\,g(X))^2$; $Y = X^2$ needs $\mathrm{E}\,X^4$.

MGF definition.

$$M_X(t) = \mathrm{E}\,e^{tX} = \sum_x e^{tx}p_X(x) \quad \text{or} \quad \int e^{tx}f_X(x)\,dx.$$

State the convergence domain. Standard moment/uniqueness theorems need finiteness on an open interval containing 0.

$$M(0) = 1, \quad M^{(r)}(0) = \mathrm{E}\,X^r,$$

$$\mathrm{Var}\,X = M''(0) - [M'(0)]^2.$$

The coefficient of t^r is $\mathrm{E}\,X^r/r!$. For $K(t) = \ln M(t)$, $K'(0) = \mu$, $K''(0) = \sigma^2$. Equal MGFs near zero imply equal distributions; equal means/variances do not.

$$M_{aX+b}(t) = e^{bt}M_X(at).$$

For independent X_i,

$$M_{\sum a_iX_i+b}(t) = e^{bt}\prod_i M_{X_i}(a_it).$$

Intersect transformed domains; without independence do not multiply marginal MGFs.

Fast forms. Rate-λ exponential: $M(t) = \lambda/(\lambda - t)$ for $t < \lambda$; mean $1/\lambda$, variance $1/\lambda^2$. If $M(t) = e^{\alpha t + \beta t^2}$, mean α, variance 2β ($\beta \geq 0$). $\int_0^\infty x^n e^{-ax}\,dx = n!/a^{n+1}$ for $a > 0$, integer $n \geq 0$.

3 Products, combinations, conditioning

Product moments.

$$\mu'_{rs} = \mathrm{E}[X^rY^s], \quad \mu_{rs} = \mathrm{E}[(X-\mu_X)^r(Y-\mu_Y)^s].$$

$$\mathrm{Cov}(X,Y) = \mathrm{E}\,XY - \mu_X\mu_Y, \quad \rho = \frac{\mathrm{Cov}(X,Y)}{\sigma_X\sigma_Y}.$$

Correlation needs positive finite variances; $|\rho| \leq 1$ and $|\,\mathrm{Cov}(X,Y)| \leq \sigma_X\sigma_Y$.

$$\mathrm{Cov}(X,Y) = \mathrm{Cov}(Y,X), \quad \mathrm{Cov}(X,X) = \mathrm{Var}\,X,$$

$$\mathrm{Cov}(aX + b, cY + d) = ac\,\mathrm{Cov}(X,Y).$$

Independence implies zero covariance (finite second moments); converse false. For $X \sim U(-1,1)$, $Y = X^2$, $\mathrm{E}\,X = \mathrm{E}\,X^3 = 0$ so covariance is zero. Yet $A = \{|X| < 1/2\}$, $B = \{Y < 1/4\}$ give $P(A \cap B) = 1/2 \neq 1/4 = P(A)P(B)$.

One combination. $W = c + \sum_i a_iX_i$:

$$\mathrm{E}\,W = c + \sum_i a_i\mu_i,$$

$$\mathrm{Var}\,W = \sum_i a_i^2\sigma_i^2 + 2\sum_{i<j} a_ia_j\sigma_{ij}.$$

Here $\sigma_{ij} = \mathrm{Cov}(X_i, X_j)$. Constants vanish in variances; preserve coefficient signs in cross terms; each unordered pair appears once, times 2. Zero covariance suffices to remove its cross term.

Two combinations. $U = c + \sum_i a_iX_i$, $V = d + \sum_i b_iX_i$:

$$\mathrm{Cov}(U,V) = \sum_i\sum_j a_ib_j\sigma_{ij}.$$

Compute both variances for ρ_{UV}. A sum over all i, j already counts both off-diagonal orders; do not insert another factor 2. Matrix shortcut is on the back.

Conditional moments. Fix the given y:

$$m(y) = \mathrm{E}[X \mid Y = y] = \frac{\int x f(x,y)\,dx}{f_Y(y)}$$

(use sums and p_Y for discrete). Replace x by $g(x)$ for $\mathrm{E}[g(X) \mid Y = y]$. Integrate only the conditional slice; denominator positive.

$$\mathrm{Var}(X \mid Y = y) = \mathrm{E}[X^2 \mid Y = y] - m(y)^2.$$

$m(y_0)$ is a number; $\mathrm{E}[X \mid Y] = m(Y)$ is a random variable.

$$\mathrm{E}\,X = \mathrm{E}[\mathrm{E}(X \mid Y)],$$

$$\mathrm{Var}\,X = \mathrm{E}[\mathrm{Var}(X \mid Y)] + \mathrm{Var}(\mathrm{E}[X \mid Y]).$$

Independence gives $\mathrm{E}[g(X) \mid Y] = \mathrm{E}\,g(X)$; a constant conditional mean alone does not prove independence. For an event B, $P(A \mid B) = P(A \cap B)/P(B)$. Continuous $Y = y$ uses a density, not division by $P(Y = y) = 0$.

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App: PRGM > STAM2 • Classic syntax below • Printed -> means STO ▶

BACK: TI-84 PLUS CE WORKFLOWS

4 Lists: most discrete calculations

1-Var Stats: let weights do the work. Enter values in L_1, nonnegative weights in L_2; at least one weight positive. Weights need not sum to 1.

```
1-Var Stats L1,L2
```

STAT > CALC; wizard: List L_1, FreqList L_2. Read $\bar{x} = \mathrm{E}\,X$, $\sigma_x^2 = \mathrm{Var}\,X$, $n = \sum w$. Use population σ_x, not sample S_x; fractional weights can leave S_x undefined.

HW5 Q1: $L_1 = \{1, 2, 3, 4\}$, $L_2 = \{2, 6, 12, 20\}$. No need to divide the list by 40 first. Output $\bar{x} = 13/4$, $\sigma_x^2 = 63/80$, $c = 1/n = 1/40$.

Functions of X. Put transformed values in another list, keeping weights:

```
L1^2->L3
1-Var Stats L3,L2
```

Read $\mathrm{E}[X^2]$ and $\mathrm{Var}(X^2)$ directly. For any g, use $g(L_1)$ in L_3. For a raw moment,

```
sum(L1^R*L2)/sum(L2)
```

For a central moment, store the mean in M, then use `sum((L1-M)^R*L2)/sum(L2)`.

Joint PMF: one row per allowed pair. L_1: every x; L_2: matching y; L_3: nonnegative joint weights. Repeat coordinates as needed. Let

```
sum(L3)->S
```

If already probabilities, $S = 1$. Otherwise normalizing constant is $1/S$. General product moment:

```
sum(L1^R*L2^T*L3)/S
```

For example $R = T = 1$ gives $\mathrm{E}\,XY$.

Events: masks select rows.

```
sum((L1>L2)*L3)/S
sum((L1+L2<=3)*L3)/S
```

These give $P(X > Y)$ and $P(X + Y \le 3)$. Use the single $\le$ token for printed <=, from [2nd][MATH] TEST.

Conditionals: mask then reuse Stats. For $X \mid Y = A$, store the given value in A:

```
L3*(L2=A)->L4
1-Var Stats L1,L4
```

Then $\bar{x}$ is the conditional mean and σ_x^2 the conditional variance. The mask's total must be positive. For a marginal mass,

```
sum((L2=A)*L3)/S
```

For $Y \mid X = A$, mask with L1=A and run Stats on L_2. For $P(X > H \mid Y = A)$ divide `sum((L1>H)*L4)` by `sum(L4)`.

5 Continuous: one reusable setup

Save the shape in Y_1. In Y=, enter nonnegative unnormalized $q(X)$; bounds B, C. For a conditional slice $q(x, y)$, let X be the integration variable and A the fixed given value; B, C must match that slice. Example HW5 Q7: $Y_1 = X + A$, $A = 1/2$, $B = 0$, $C = 1$.

```
fnInt(Y1,X,B,C)->D
fnInt(X*Y1,X,B,C)/D->M
fnInt((X-M)^2*Y1,X,B,C)/D
```

These give mass D, mean M, and variance. Normalization cancels in the ratios. For a full unnormalized density, $c = 1/D$; for a joint slice, D is the appropriate marginal value if the joint is already normalized.

Any moment or payoff.

```
fnInt(X^R*Y1,X,B,C)/D
```

Replace X^R by $g(X)$ for $\mathrm{E}\,g(X)$. To get $\mathrm{Var}(X^2)$, first store

```
fnInt(X^2*Y1,X,B,C)/D->M
```

then evaluate

```
fnInt((X^2-M)^2*Y1,X,B,C)/D
```

The centered form avoids subtracting two nearly equal large moments. HW5 Q2: $Y_1 = 1 - X^2$, $B = -1$, $C = 1$ gives $\mathrm{Var}\,X = 1/5$, $\mathrm{E}\,X^2 = 1/5$, $\mathrm{Var}(X^2) = 8/175$.

Probability by ratio. For $[E, F]$ within $[B, C]$,

```
fnInt(Y1,X,E,F)/D
```

For $P(X > E)$, integrate from E to C; clip bounds to the support. For HW4 Q2 conditional $X \mid Y = A$, use $Y_1 = 2X + A$, $B = 0, C = 1$. For each new A, recompute D, M.

Double integrals. fnInt cannot be nested. Draw the region and do one integral by hand, then put the remaining function in Y_1. HW4 Q5 reduces to

```
fnInt(1/2+X-5*X^2/8,X,0,1)
```

which gives 19/24. Split a changing boundary before calculating. A one-dimensional marginal at a fixed value can use fnInt directly; a symbolic marginal formula still needs algebra.

MGF value from a PDF. With T a specified number,

```
fnInt(e^(T*X)*Y1,X,B,C)/D
```

computes $M(T)$ on a finite support. It does not establish convergence on an infinite support or provide a symbolic MGF. Avoid using arbitrary large finite bounds as if they were ∞.

6 Covariance, matrices, MGFs, keys

2-Var Stats from joint rows. Using $L_1 = x, L_2 = y, L_3 = w$ from column 4,

```
2-Var Stats L1,L2,L3
```

read $\bar{x}, \bar{y}, \sigma_x, \sigma_y, n, \Sigma xy$:

$$\mathrm{Cov}(X, Y) = \frac{\Sigma xy}{n} - \bar{x}\bar{y}, \quad \rho = \frac{\mathrm{Cov}(X, Y)}{\sigma_x \sigma_y}.$$

Get actual statistics tokens through VARS > Statistics. Store results before editing lists or running another Stats command. Direct list alternative:

```
sum(L1*L3)/S->M
sum(L2*L3)/S->N
sum((L1-M)*(L2-N)*L3)/S
```

This is covariance, with $S =$ sum(L3).

Matrices: automate every cross term. [2nd][x^{-1}] MATRIX > EDIT. Enter covariance matrix $[A] = \Sigma$ with variances on its diagonal. Put coefficients of one or more combinations in rows of $[B]$. Then

$$[B][A][B]^{\mathsf{T}}$$

is their covariance matrix. One row gives a 1×1 variance. Two rows give $\begin{pmatrix} \mathrm{Var}\,U & \mathrm{Cov}(U, V) \\ \mathrm{Cov}(U, V) & \mathrm{Var}\,V \end{pmatrix}$. Ignore additive constants. The transpose is the actual postfix T token from MATRIX > MATH, not alpha-T. HW5 Q6:

$$[A] = \begin{pmatrix} 4 & 1 & -1 \\ 1 & 9 & 0 \\ -1 & 0 & 2 \end{pmatrix}, \quad [B] = \begin{pmatrix} 3 & -2 & 1 \end{pmatrix}.$$

Type $[B] * [A] * [B]^{\mathsf{T}}$ to obtain [[56]]. If given correlation, first use $\mathrm{Cov}(X, Y) = \rho \sigma_X \sigma_Y$.

A given MGF in Y_1. Enter $M_X(X)$; X now represents t. For $Z = aX + b$, store a in A, b in B; put this transform in Y_2:

```
e^(B*X)*Y1(A*X)
```

Use actual Y_1 token from VARS > Y-VARS > Function. The mean is checked by

```
nDeriv(Y1,X,0)
```

Derive $M''(0)$ by hand, or recognize $e^{\alpha t + \beta t^2}$: mean α, variance 2β. Do not use nested numerical derivatives as exact moments.

Fast entry and reliable use. [2nd][0]: CATALOG. [2nd][STAT]: LIST; seq under OPS, sum under MATH. A finite sum is

```
sum(seq(expression,K,first,last))
```

Store reusable numbers with STO ▶; do not rely on Ans after another command. MATH > Frac suggests a fraction, not a proof. Calculator output handles arithmetic; write the support, event, formula, bounds, MGF domain, and independence justification yourself.